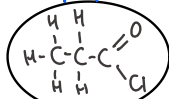
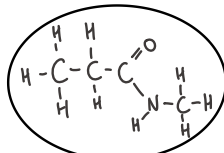
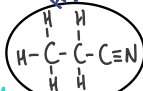
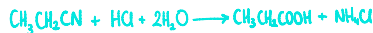
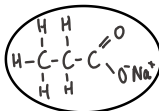
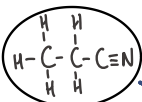
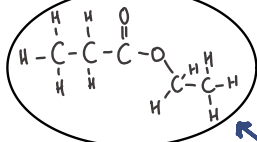
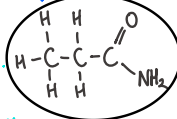
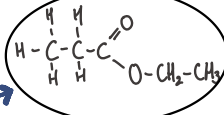
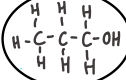
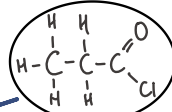
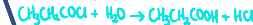
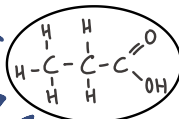


UNIT 4

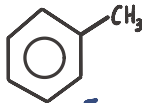
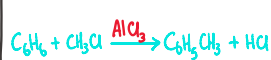


- Halogenation
- React with PCl_5 in anhydrous conditions.

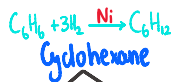


UNIT 5

Topic 18

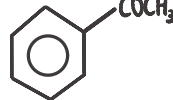


Methylbenzene.



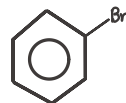
Hydrogenation.
Catalyst: Nickel.

Phenylethanone



Acylation.

Bromobenzene.



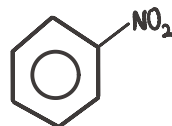
Bromination.

- Reaction carried out under UV light.
- Benzene heated with Br_2 under reflux AND in the presence of a halogen-carrier. eg: AlCl_3 , AlBr_3 , FeBr_3 , Fe filings.



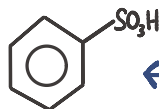
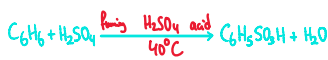
Benzene

Nitration.



Nitrobenzene.

- Benzene reacts with NO_2
- Nitro group (NO_2) is made by the reaction of mixture of conc. HNO_3 & H_2SO_4 .
$$\text{H}_2\text{SO}_4 + \text{HNO}_3 \longrightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$$
- Carried out at $50-60^\circ\text{C}$.
- Higher temp. can lead to formation of dinitrobenzene/trinitrobenzene.



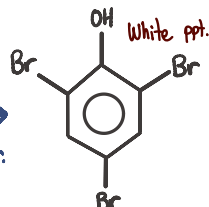
Sulfonation.

- Carried out with fuming sulfuric acid at 40°C .
- Fuming H_2SO_4 is made by dissolving sulfur(VI) oxide (sulfur trioxide) in concentrated H_2SO_4 .



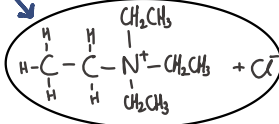
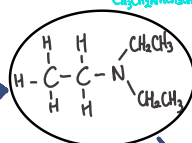
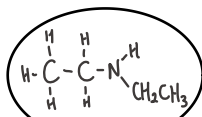
Bromination of Phenol

- Substitution.
- Reacts with Br_2 water.
- Decolorizes Br_2 water.



2,4,6-tribromophenol

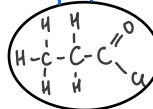
UNIT 5 TOPIC 19



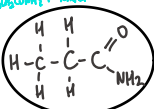
Reaction with Halogenoalkane.



Propionyl Chloride.

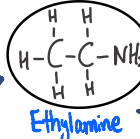
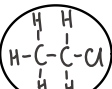


Preparation of Amides.
Reacts with conc. $NH_3(aq)$



Propionamide.

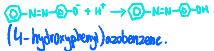
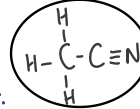
Preparation of Amine from Haloalkane.



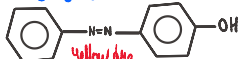
Ethylamine

Preparation from nitriles.

Reagents: Tetrahydraluminate ($LiAlH_4$) + dry ether.



(4-hydroxyphenyl)azobenzene.

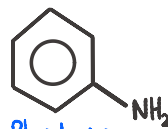


Yellow dye.



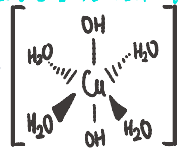
Nitrobenzene

Reagent: Tin mixed with conc. HCl heated under reflux.
Reduction from Nitrobenzene to phenylamine.



Phenylamine

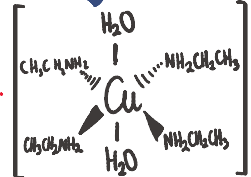
Reaction with Copper(II) ions.



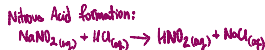
Pale Blue ppt.



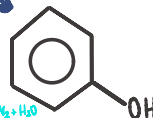
Deep blue solution.



Reaction with Nitrous Acid at $T \leq 5^\circ C$.



Nitrous Acid formation:



Phenol

Reaction with Nitrous Acid at $T > 5^\circ C$.



Benzediazonium ion.



Phenoxide ion.

